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Tertiary paleomagnetism of North and South China and a reappraisal of late Mesozoic paleomagnetic data from Eurasia: implications for the Cenozoic tectonic history of Asia

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Abstract

To contribute to the understanding of the Cenozoic tectonic history of China, and eastern Eurasia in general, we collected paleomagnetic samples from Tertiary rocks in both North and South China. Our Miocene paleomagnetic data for northern China are from four widely separated basalt sections in the Inner Mongolia Autonomous Region and Shandong Province. Both normal and reversed polarities of the characteristic magnetization are present in the stratigraphic sequence, and the mean normal and reversed directions are closely antiparallel to each other. The fact that the same characteristic direction was carried by both magnetite and hematite suggests that the hematite was probably formed by auto-oxidation processes during primary cooling of the basalts. The paleomagnetic data from these four areas also pass a regional generalized fold test. The corresponding mean Miocene paleomagnetic pole for northern China is longitude 261.6°E, latitude 88.3°N ($A_{95} = 6.4°$). Our Early Tertiary paleomagnetic results for south China are from two sandstone sections around the city of Nanning (22.7°N, 108.3°E) in Guangxi Zhuang Autonomous Region. High-temperature characteristic directions isolated in 31 samples from five sites pass both tilt and reversal tests. The paleomagnetic pole position determined from the five sites lies at longitude 228.0°E, Latitude 83.6°N ($A_{95} = 8.5°$).

The Tertiary paleomagnetic poles for North and South China are statistically indistinguishable from each other and thus confirm that North and South China were fully assembled at least by the Tertiary. Our new Tertiary paleomagnetic data are consistent with the hypothesis that the major continental blocks of North and South China were in the same position as today during the Tertiary. A critical evaluation and selection of paleomagnetic results from the Late Jurassic onward for Siberia, Europe and North and South China further corroborate this hypothesis. Most previously existing Jurassic and Cretaceous paleomagnetic poles from Europe/Siberia are of uncertain reliability and should not be used for tectonic reconstruction. Using poles derived from transferring reliable data from other continents, we show that the previous paleomagnetic inferences of large eastward displacements between these Chinese blocks during the Cenozoic are no longer warranted. The available paleomagnetic data that meet our acceptance criteria are also of interest concerning the boundary conditions of the propagating extrusion model for Asian tectonics. Additional Mesozoic and Cenozoic paleomagnetic data from Siberia and North and South China

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are still needed to resolve the smaller, second-order displacements that appear to characterize the Cenozoic tectonic history of this part of Asia.

1. Introduction

Eastern Eurasia makes up a broad region of active plate tectonics and is a natural laboratory for studying accretion of crustal blocks and intracontinental deformation (Fig. 1). It is now generally recognized that this part of Eurasia is formed by the accretion of three major cratonic blocks: Siberia (SIB) and the North and South China Blocks (NCB and SCB, respectively). The time of final suturing of these blocks, although not precisely determined, is certainly not later than the late Mesozoic (e.g., Klimetz, 1983; Zhang et al., 1984; Courtillot and Besse, 1986). Since the time of suturing, however, there has been significant Cenozoic tectonic activity. To the east, subduction of the Pacific plate to Asia, beginning in the Jurassic–Cretaceous and continuing into the Cenozoic, climaxed with collision of the Sikhote Alin block, the opening of the Japan Sea, and formation of the Ryukyu islands (e.g., Jolivet et al., 1988; Kimura et al., 1990). To the west, the collision of India with Asia, which has been regarded as Eocene in age (about 50 Ma, see Besse et al., 1984) but which recent studies suggest collision may have begun as early as the Cretaceous–Tertiary boundary (about 65 Ma, see Jaeger et al., 1989), continues to the present. Besides the formation of the Himalayas and the uplift of the Tibetan plateau, the continued penetration of Asia by India appears to be causing large-scale intracontinental deformation and crustal block rotations and/or translations in eastern Eurasia, as suggested by many workers (Molnar and Tapponnier, 1975; Tapponnier and Molnar, 1977; 1979; Tapponnier et al., 1982; 1990; Tapponnier, 1988; Peltzer and Tapponnier, 1988).

Paleomagnetism remains one of the best tools for testing and quantifying the relative displacement of crustal blocks. In recent years considerable amounts of pre-Cenozoic paleomagnetic data from both North and South China have been reported (e.g., McElhinny et al., 1981; Chan et al., 1984; Lin et al., 1985; Kent et al., 1986; 1987; Opdyke et al., 1986; Zhao and Coe, 1987; 1989; C.Z. Tan and Zhai, 1988; W. Fang et al., 1989; Enkin et al., 1991a,b; Huang et al., 1992a; Zhao et al., 1992). However, very few attempts at paleomagnetic studies on Cenozoic rocks have been made. By comparing Mesozoic paleomagnetic APW paths for the North and South China blocks, Tarim and Europe/Siberia, several workers have concluded that the Chinese blocks have been pushed eastward more than 2000 km relative to Siberia, mainly due to the India–Asia collision (Lin and Fuller, 1986; 1990; Li, 1990). In contrast, other workers have questioned whether the

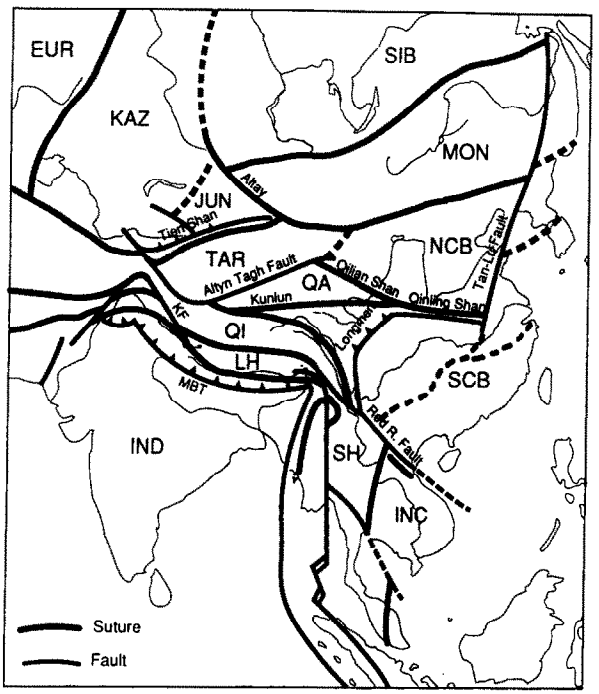


Fig. 1. Sketch map of present location of continental blocks of China and Southeast Asia showing main sutures and faults. *KF* = Kharakorum Fault; *MBT* = Main Boundary Fault. The major blocks are: *EUR* = Europe; *KAZ* = Kazakhstan; *SIB* = Siberia; *MON* = Mongolia; *JUN* = Junggar; *TAR* = Tarim; *NCB* = North China Block; *SCB* = South China Block; *QA* = Qaidam; *QI* = Qiangtang (North Tibet); *LH* = Lhasa (South Tibet); *SH* = Shan Thai; *INC* = Indochina; and *IND* = India. Modified after Enkin et al. (1991a).

Jurassic and Cretaceous paleomagnetic poles for Europe/Siberia are sufficiently well determined to draw such conclusions (Courtillot and Besse, 1986; Lee et al., 1987; Kim and Van der Voo, 1990; Zhao et al., 1990a,b; Besse and Courtillot, 1991).

In this paper we first present our own Tertiary paleomagnetic results from four sampling areas in North and South China. We then evaluate and compare other coeval paleomagnetic results from China. We subsequently assess the main body of paleomagnetic results from the Late Jurassic onward for Siberia and North and South China. Based on a selection of these data, we discuss the first-order paleomagnetic implications concerning the Cenozoic tectonic history of Eastern Eurasia. The data selection criteria that we used were quite lenient, and similar to those generally accepted (e.g., Van der Voo, 1990; Besse and Courtillot, 1991), with an emphasis on evidence for a primary magnetization through standard demagnetization techniques, reasonably accurate age dating (to the stage level for geologic periods), presence of reversals and other stability tests that constrain the age of magnetization (e.g., contact, conglomerate, fold, or tilt tests), and analysis for appropriate tectonic coherence.

2. Geological setting and paleomagnetic sampling

2.1. North China Block

Our paleomagnetic sampling in the North China Block was confined to Shandong Province (Fig. 2a). As we will often refer to province names from which samples were taken, we provide a political map of China in Fig. 2b. The geologic setting of Shandong Province is rather complicated. The famous Tancheng–Lujiang fault zone runs across this province and separates it into eastern and western parts. Paleozoic sedimentary rocks are abundant in western Shandong, but are completely absent in eastern Shandong. Quaternary basaltic eruptions are very common in eastern Shandong, whereas essentially none occur in western Shandong. Early Cretaceous and Miocene volcanic rocks are distributed in both western and eastern parts of Shandong.

We sampled two Miocene localities with basaltic lava flows in the Yishui county of western Shandong. At one locality (Hongshiya: 36.2°N, 118.5°E, hereafter named the first locality), 28 samples were collected from eleven flows that spanned about 100 m. At another locality (Majiaowang: 36.4°N, 118.8°E, termed the second locality) only five additional samples were obtained from five flows due to sampling difficulties during severe snowstorms. At both localities the basalt flows occur as essentially horizontal layers and crop out along mountain tops. The flat-lying occurrence of the basalt flows supports the geologic conclusion that there has been no significant tectonic deformation in this region since the late Tertiary basalts erupted. These basalts are reported to contain xenoliths of lherzolite and yield K–Ar isotopic ages between 14 and 18 Ma (Geological Society of Shandong, 1982).

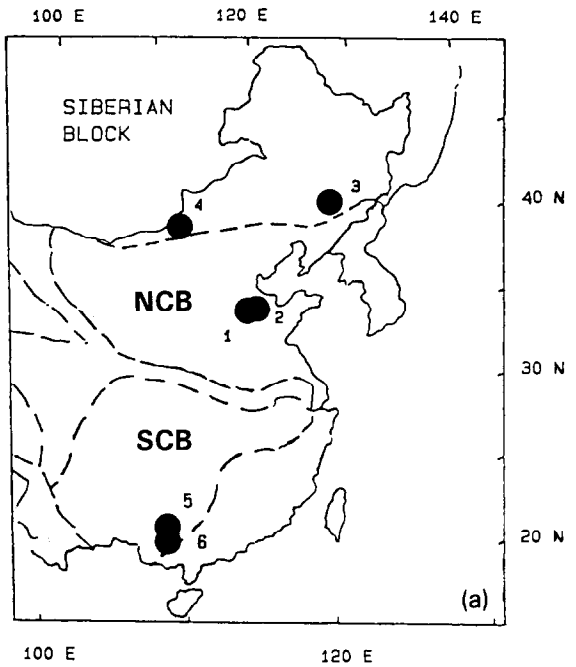


Fig. 2. (a) Paleomagnetic sampling localities of this study in North and South China. *1* = Honshiya; *2* = Majiaowang; *3* = Pingzhuang; *4* = Jining; *5* = Qingxiushan; *6* = Wuxianling. (b) The political provinces of China.

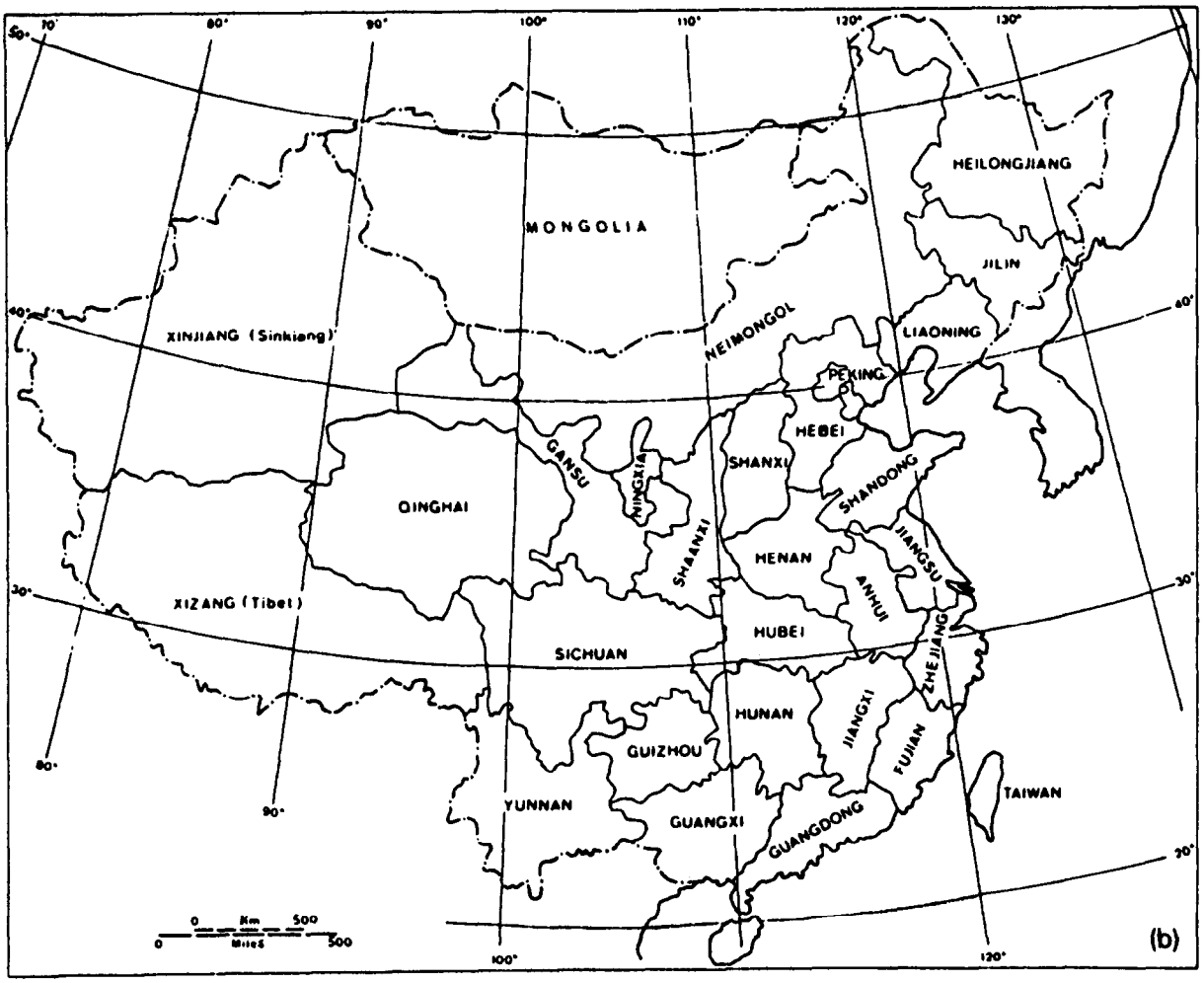


Fig. 2. Continued.

2.2. Inner Mongolia

Inner Mongolia is a place of particular interest for plate tectonics. It is located in the area between the NCB and Siberia and is proposed to be the location of the suture zone between the two blocks (Quan, 1983). In the Pingzhuang area (42.0°N, 119.2°E) of eastern Inner Mongolia, we sampled a late Tertiary basalt formation that crops out well at two localities. At the first locality columnar basalt flows occur as horizontal layers, while at the second locality flows were tilted to the east with enough structural variation to enable a definitive tilt test to be made. Unfortunately, both the age of tilting and the isotopic ages for these basalts are not available. The formation is dated as late Tertiary, largely on the basis of plant fossils identified by local geologists (Lin Yu, pers. commun., 1987). A total of 63 samples was taken from six sites, representing at least six flows. About 650 km to the southwest of the Pingzhuang area, we sampled another Miocene basalt formation, the famous Hannuoba Basalt Formation, in the Jining area (40.9°N, 113.2°E) of central Inner Mongolia. K–Ar ages range from 14.1 to 25.4 Ma (Deng et al., 1988; Liu, 1988). The flows are 3–10 m thick, on average, although flows as thick as 20 m are encoun-

tered. The structure of the Hannuoba Basalt Formation in the Jining area is simple, with flows generally occurring as horizontal layers. No major faults are observed to duplicate the section. Forty-seven oriented samples were collected from seven flows in the Jining area.

2.3. South China Block

The South China Block defined in this study comprises two smaller blocks—the Yangtze block and the Huanan block (Hsü et al., 1988; Rowley et al., 1989). Although estimates on the timing of accretion of the two blocks are still controversial (ranging from Precambrian to Mesozoic; e.g., Hsü et al., 1988; Rowley et al., 1989), it is widely agreed that the two blocks were certainly accreted by the late Mesozoic (Zhang et al., 1984; Dobson and Heller, 1992).

We sampled the early Tertiary Fenghuangshan Formation at two separate localities near the city of Nanning in Guangxi Zhuang Autonomous Region (Fig. 1). The first locality is in Qingxiushan, about 7 km east of Nanning (22.9°N, 108.4°E). The rocks sampled mostly consist of yellowish sandstones with a thickness of over 300 m. Bedding attitudes at this sampling locality were essentially flat-lying. The second locality is located at Wuxianling, about 4 km southwest of Nanning. The Fenghuangshan Formation there is composed of purplish-red sandstones and siltstones with a total thickness of less than 50 m. Strata are gently tilted northward with dips of 15° at Wuxianling, providing a weak, between-locality tilt test. A total of 32 samples were collected from five sites distributed between these two localities. Fossils found in the Fenghuangshan Formation include *Castanea miomollissima* and *Mallotus nanningensis*, which are indicators of middle Eocene age (Bureau of Geology and Mineral Resources of Guangxi Zhuang Autonomous Region, 1985).

Sampling followed standard paleomagnetic practice: drilling cores with a portable gasoline-powered drill and orientating using a magnetic compass mounted on an orienting device. We corrected for the effect of local magnetic anomalies on magnetic orientation by taking solar azimuths or by using azimuths of landmarks identifiable on maps.

3. Paleomagnetic results

Samples were measured on a Schonstedt spinner magnetometer and a 2G cryogenic magnetometer at the paleomagnetic laboratory of the University of California, Santa Cruz. Demagnetization experiments were carried out using Schonstedt equipment in a magnetically shielded room. Magnetization directions were determined by principal component analysis (Kirschvink, 1980). The distribution of paleomagnetic directions at each site were calculated using Fisher (1953) statistics, and site mean directions of all demagnetized data were derived by giving unit weight to each mean sample direction. The paleomagnetic data are summarized in Tables 1 and 2 and discussed below.

3.1. Shandong (NCB)

The initial natural remanent magnetization (NRM) intensities of the basalts at the first locality (Hongshiya) were typically of the order of 1 A/M. Thermal, alternating field (AF), and combined (thermal after AF) demagnetizations were conducted, in an attempt to remove possible isothermal remanent magnetization (IRM) caused by lightning strikes. After removal of a low-unblocking temperature component between 100° and 300°C, the samples yielded a well-defined characteristic magnetization, as shown by demagnetization plots that almost all reached a stable end point (Fig. 3a). Combined demagnetization and isothermal remanent acquisition experiments suggest that the characteristic magnetization is probably carried by magnetite. Both normal and reversed polarities of the characteristic direction are present in the stratigraphic sequence. In addition, an intermediate direction is observed in samples from the middle portion of the same stratigraphic section. These samples are overlain by lavas of reversed polarity and underlain by lavas of normal polarity, suggesting that lava flows at this locality span a polarity transition. For the

Table 1
Paleomagnetic results for Tertiary rocks sampled in North and South China from this study

Study area		Strike/dip	Polarity ratio R/N	Temp. range	Statistical analysis								
Slat. (°N)	Slong. (°E)				Directions					Poles			
					N/No	D	I	α_{95}	k	Long.	Lat.	A_{95}	K
Shandong													
1. Hongshiya locality													
36.2	118.5	0/0	0/2	400–600	2/2	10.5	60.7	–	–	170.6	80.1	–	–
36.2	118.5	0/0	0/3	400–600	3/3	18.3	60.7	4.9	641.9	181.7	74.7	6.8	326.7
36.2	118.5	0/0	0/3	400–600	3/3	16.4	63.0	6.2	391.1	169.9	75.0	9.3	177.5
36.2	118.5	0/0	2/0	400–600	2/2	192.6	−46.9	–	–	60.6	−76.4	–	–
36.2	118.5	0/0	3/0	400–600	3/3	190.5	−49.2	41.1	10.0	59.4	−79.2	54.8	6.1
36.2	118.5	0/0	2/0	400–600	2/2	168.8	−50.5	–	–	184.6	−79.3	–	–
36.2	118.5	0/0	2/0	400–600	2/2	180.2	−41.9	–	–	117.8	−78.0	–	–
36.2	118.5	0/0	2/0	400–600	2/2	181.7	−46.7	–	–	107.8	−81.6	–	–
36.2	118.5	0/0	1/0	400–600	1/1	180.5	−49.9	–	–	114.0	−84.5	–	–
Mean of normal sites (flows) $N = 3$						15.0	61.5	3.6	1170.1	174.8	76.8	5.2	564.6
Mean of reversed sites $N = 6$						182.5	−47.8	5.4	156.6	102.0	−82.4	6.5	106.6
All sites $N = 9$						5.8	52.5	6.0	75.2	233.5	84.3	6.8	57.8
2. Majiaowang locality													
36.4	118.8	0/0	1/0	425–575	1/1	194.6	−49.8	–	–	49.7	−76.6	–	–
36.4	118.8	0/0	1/0	425–575	1/1	178.8	−54.2	–	–	151.1	−88.2	–	–
36.4	118.8	0/0	1/0	425–575	1/1	170.4	−54.6	–	–	203.1	−82.1	–	–
All sites $N = 3$						181.7	−53.3	12.1	104.5	92.4	−87.2	16.3	58.5
Inner Mongolia													
3. Pingzhuang locality													
42.0	119.2	198/20	0/8	350–525	8/8	63.2	48.5	3.7	229.7	201.4	38.4	3.7	229.7
42.0	119.2	0/0	0/8	300–550	8/8	350.9	46.7	13.1	18.7	333.5	74.1	16.5	12.2
42.0	119.2	337/23	0/7	375–550	7/7	327.8	55.9	25.9	6.4	26.0	67.7	31.5	4.6
42.0	119.2	21/27	0/5	400–500	5/5	338.9	35.1	17.5	20.0	344.3	61.6	17.0	21.2
42.0	119.2	50/56	10/0	475–630	10/10	156.2	−13.7	20.2	6.7	157.1	−49.9	14.2	12.5
42.0	119.2	37/46	11/0	475–650	11/11	157.2	−14.1	30.3	3.2	154.6	−48.8	26.4	3.9
Mean of normal sites $N = 4$ (Geog. coord.)													
Mean of normal sites $N = 4$ (Strat. coord.)													

						357.1	52.4	34.2	8.2	310.3	81.8	44.7	5.2
						5.5	53.9	15.5	36.2	262.7	81.9	21.4	19.4
Mean of reversed sites $N = 2$ (Geog. coord.)						156.7	−13.9	–	–	155.8	−49.4		
Mean of reversed sites $N = 2$ (Strat. coord)						177.7	−58.7	–	–	188.1	−88.4		
All sites $N = 6$ (Geog. coord.)						347.6	39.4	27.5	7.1	329.0	70.3	28.9	6.3
All sites $N = 6$ (Strat. coord.)						3.1	55.6	9.8	47.9	268.6	84.8	13.1	27.0
4. Jining locality													
40.9	113.2	0/0	0/7	575–675	7/7	345.6	55.3	4.9	155.0	4.0	77.7	6.4	89.6
40.9	113.2	0/0	0/12	575–625	12/12	358.0	53.3	2.0	352.1	305.9	82.8	2.6	212.1
40.9	113.2	0/0	0/6	585–675	6/6	354.0	48.9	5.4	153.9	320.1	78.1	5.1	176.7
40.9	113.2	0/0	0/5	525–675	5/5	357.5	54.6	8.1	89.5	314.1	84.6	9.2	70.5
40.9	113.2	0/0	5/0	525–585	5/5	180.3	−68.4	4.8	254.0	293.3	−79.1	7.7	98.5
40.9	113.2	0/0	5/0	575–625	5/5	162.6	−76.2	8.5	81.3	275.3	−64.4	15.2	26.3
40.9	113.2	0/0	6/0	615–675	6/6	173.8	−75.9	6.5	107.8	286.4	−66.8	11.2	36.8
Mean of normal sites $N = 4$						353.9	53.1	5.0	335.0	330.1	81.5	6.3	214.1
Mean of reversed sites $N = 3$						173.5	−73.2	7.4	277.3	282.9	−70.4	12.5	96.7
All sites $N = 7$						353.8	61.7	8.7	49.5	67.3	83.8	11.5	28.5
Guangxi													
5. Qingxiushan locality													
22.9	108.4	0/0	2/3	600–650	5/6	358.8	37.5	9.9	60.4	315.4	88.2	9.1	72.4
22.9	108.4	0/0	1/6	625–660	7/7	3.2	26.0	5.1	142.0	269.4	80.5	4.7	164.0
22.9	108.4	0/0	3/5	625–660	8/8	1.0	30.0	7.3	59.0	278.8	83.6	6.1	83.9
6. Wuxianling locality													
22.7	108.4	234/15	6/0	600–680	6/6	186.5	−39.9	10.7	40.1	15.3	−84.0	11.7	33.8
22.7	108.4	234/15	5/0	600–685	5/5	200.1	−40.3	16.2	23.2	13.3	−71.2	19.3	16.7
Mean of normal sites $N = 3$ (Strat. coord.)						1.1	31.2	9.4	174.9	278.5	84.2	6.5	358.1
Mean of reversed sites $N = 2$ (Geog. coord.)						205.8	−48.7	–	–	355.9	−66.0	–	–
(Strati. coord)						193.4	−40.3	–	–	14.6	−77.6	–	–
All sites $N = 5$ (Geog. coord.)						10.1	39.8	14.2	29.8	193.5	80.0	14.2	30.1
(Strati. coord.)						5.7	35.0	8.9	75.3	228.0	83.6	8.5	81.1

Slat(°N), Slong(°E) = site latitude and longitude.
Polarity ratio = number of reversed (R) to number of normal (N) samples.
Temp. range = range of unblocking temperatures of characteristic component during thermal demagnetization.
No: Number of samples exhibiting the characteristic component.
N: Number of samples used in statistical analysis.
D = mean declination in degrees.
I = mean inclination in degrees (*directions corrected for tilt*).
k = Fisher precision parameter for mean direction.
α_{95} = Radius (degrees) of circle of 95% confidence about the mean direction.
Long = East longitude of Virtual Geomagnetic Pole (VGP).
Lat = North latitude of VGP calculated from *corrected mean directions*.
K = Fisher precision parameter for the pole.
A_{95}: Radius of circle of 95% confidence about the VGP.

purpose of this study, we eliminated the transitional directions, which are defined by their virtual geomagnetic poles (VGP's) falling more than 45° from both the mean normal and reversed VGP for the entire data set, and by the surrounding sites that are of full polarity. The mean direction for both normal and reversed flows is $D = 5.8°$, $I = 52.5°$, $N = 9$ flows, $\alpha_{95} = 6.0°$, and $k = 75.2$. The corresponding mean VGP is at Long = 233.5°E, Lat = 84.3°N, $A_{95} = 6.8°$, and $K = 57.8$ (Table 1).

Paleomagnetic data from samples at the second locality (Majiaowang), however, showed an important difference. The NRM of the samples is dominated by a direction with northwesterly declinations and negative inclinations (Fig. 3b). A soft overprint with shallow inclination collapses rapidly by 20 mT AF demagnetization, indicating an IRM origin caused by lightning. Further demagnetization revealed the characteristic direction (reversed polarity) at higher demagnetization steps. The mean is $D = 181.7°$, $I = -53.3°$, $N = 3$ samples, $\alpha_{95} = 12.1°$, $k = 104.5$, and is almost identical to the mean for flows in the first locality, even though only three flows are represented.

The combined mean direction for rocks at

Table 2
Grand mean of normal and reversed characteristic directions and mean of localities

	P	n/No	C	Direction				Pole position			
				D	I	α_{95}	k	Long.	Lat.	A_{95}	K
North China Block (Shandong)											
Averaging by sites											
Normal	N	3/12	G/S	15.0	61.5	3.6	1170	174.8	76.8	5.2	237.9
Reversed	R	9/12	G/S	182.2	−49.6	4.5	132.5	100.5	−84.1	5.4	90.6
Mixed	M	12/12	G/S	4.8	52.7	4.7	86.1	*238.4*	*85.2*	*5.6*	61.9
Averaging by locality											
$N = 2$			G/S	3.8	52.9	–	–	246.1	86.0	–	–
Mongolian Block (Inner Mongolia)											
Averaging by site											
Normal	N	8/13	G	355.4	52.8	13.2	18.6	321.2	81.8	16.8	11.8
			S	359.6	53.7	6.9	65.9	298.0	83.1	9.3	36.1
Reversed	R	5/13	G	161.8	−50.5	33.8	6.1	204.6	−76.2	30.4	7.3
			S	175.8	−67.4	9.4	67.1	279.7	−78.3	12.7	37.3
Mixed	M	13/13	G	350.3	52.2	12.9	11.2	352.9	81.2	13.8	9.9
			S	358.5	59.0	6.1	47.0	*29.2*	*88.6*	*8.2*	26.3
Averaging by locality											
$N = 2$			G	350.0	50.6	–	–	347.4	80.1	–	–
			S	358.9	58.7	–	–	12.9	88.8	–	–
South China Block (Guangxi)											
Averaging by samples											
Normal	N	14/32	G/S	1.7	28.9	4.9	65.9	275.7	82.7	3.9	102.8
Reversed	R	17/32	G	195.4	−44.9	7.4	24.0	358.7	−75.3	8.6	18.3
			S	188.1	−38.2	6.4	32.0	25.2	−82.3	7.1	26.1
Mixed	M	31/32	G	8.7	38.2	5.5	23.3	200.7	81.5	5.5	22.8
			S	5.0	34.0	4.3	36.3	*236.0*	*83.8*	*4.3*	36.3
Averaging by locality											
$N = 2$			S	6.9	35.9	–	–	218.6	82.9	–	–

P = polarity; N = normal; R = reversed, M = both N and R; n/No = number of samples, sites, localities, or areas used in statistical analysis/total of the samples, sites, localities, or areas exhibiting the characteristic component; C = coordinates; G = Geographic; S = Stratigraphic.
See Table 1 for explanation of other abbreviations.

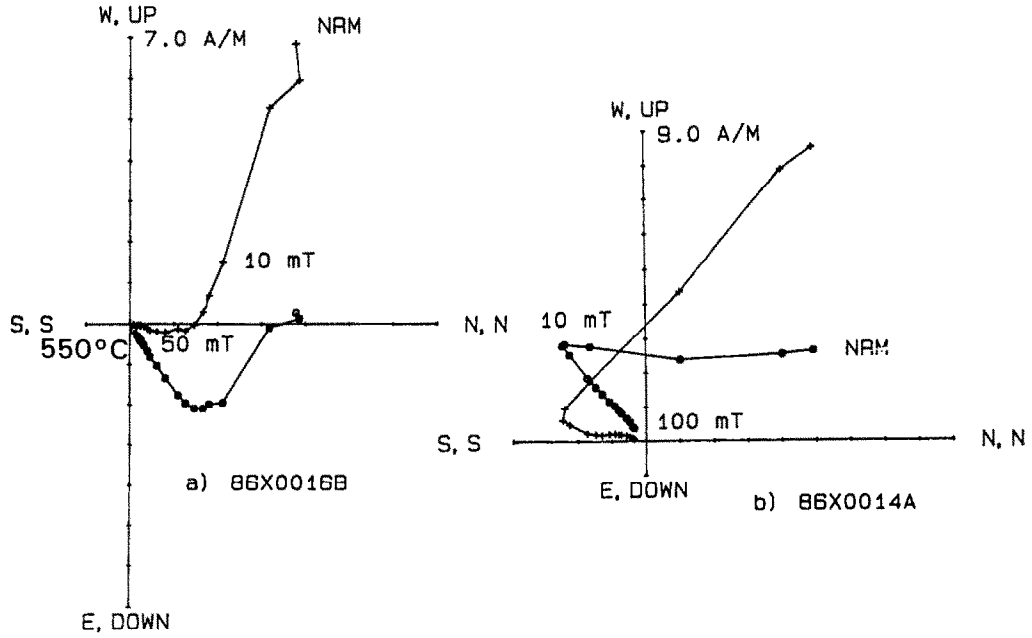


Fig. 3. Representative vector plots of Miocene basalts from Shandong. (a) Combined (thermal after AF) demagnetization of sample 86X0016B from the Hongshiya area. (b) AF demagnetization of sample 86X0014A from the Majiaowang area. Directions are plotted in geographic coordinates. Crosses = vector endpoints projected onto the horizontal plane; circles = endpoints projected onto the north–south vertical plane; NRM = natural remanent magnetization.

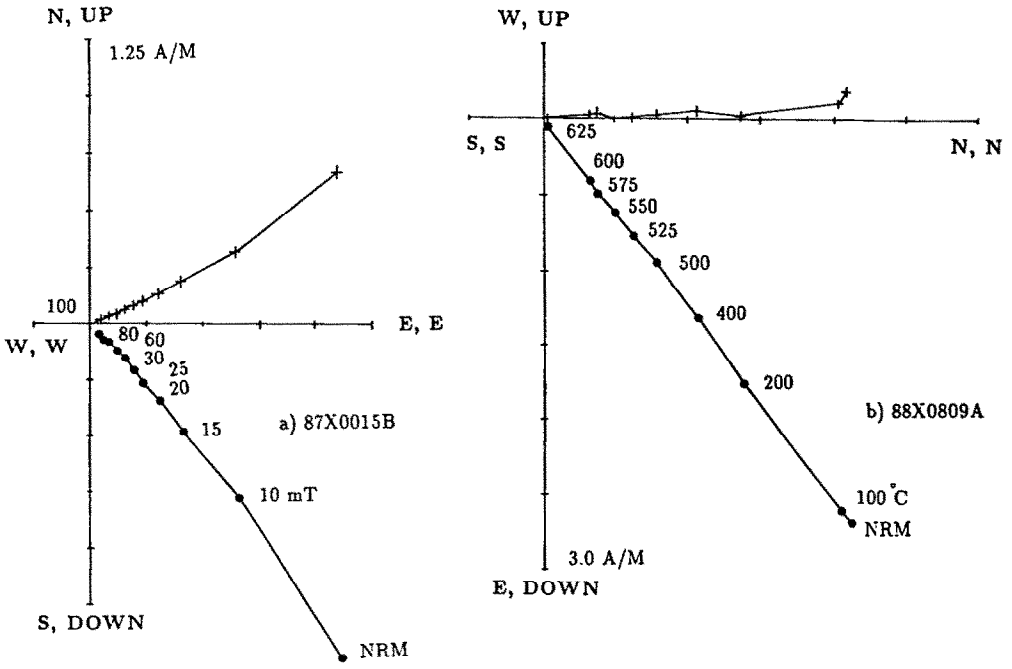


Fig. 4. Representative vector plots of Tertiary basalts from Inner Mongolia. (a) AF demagnetization of sample 87X0015B from the Pingzhuang area. (b) Thermal demagnetization of sample 88X0809A from the Jining area. Directions are plotted in geographic coordinates. Crosses and circles as Fig. 3.

Table 3
F-test on Tertiary paleomagnetic poles for Chinese blocks

Study	N	R	Long.	Lat.	A_{95}	K
Mongolian Block (Inner Mongolia)						
This study	13	12.55	29.2	88.6	8.2	26.3
Zheng et al. (1991)	49	44.9	192.8	83.4	6.2	11.7
Average Zheng et al. (1991) and this study (locality mean in Table 1)						
N = 3 [b] localities		*2.98*	*113.4*	*86.7*	*9.2*	*181.3*
North China Block (Shandong)						
This study	12	11.82	238.1	85.2	5.6	61.9
Lin et al. (1985)	4	3.98	177.2	76.4	7.7	143.6
Average Lin et al. (1985) and this study (locality mean in Table 1)						
N = 3 localities		*2.98*	*201.9*	*83.9*	*11.6*	*114.0*
South China Block (Guangxi)						
This study	31	30.19	236.0	83.8	4.3	36.3
Liang et al. (1986)	28	24.79	325.2	84.4	~ 10 [a]	8.4
Average Liang et al. (1986) and this study (locality mean in Table 1)						
N = 3 [b] localities		*2.97*	*241.7*	*85.4*	*13.5*	*84.5*

both localities is $D = 4.8°$, $I = 52.7°$, $N = 12$ flows, $\alpha_{95} = 4.7°$, $k = 86.1$. The corresponding pole position is Long = 238.4°E, Lat = 85.2°N, $A_{95} = 5.6°$, $K = 61.9$ (Table 2).

3.2. Inner Mongolia

3.2.1. Pingzhuang area

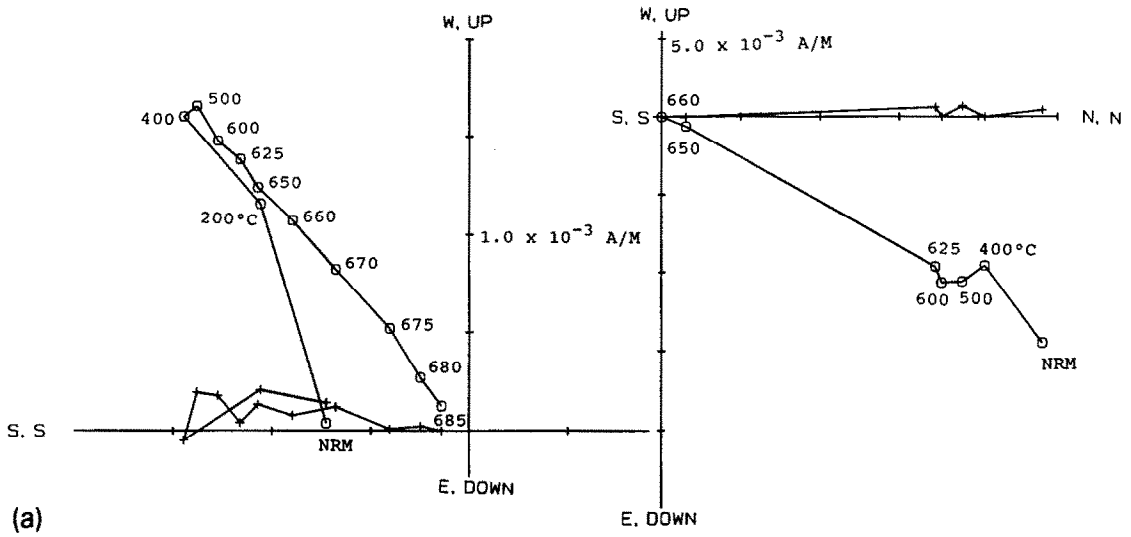
We performed stepwise thermal and AF demagnetization experiments on 63 samples from six sites (covering at least six lava flows). The magnetization of Tertiary basalt samples from the Pingzhuang area is quite stable and well behaved. A small, random component of magnetization was easily removed by demagnetization and the characteristic component quickly revealed (Fig. 4a). Normal directions were obtained from four flows, and reversed directions were isolated in two flows, respectively (Table 1). The overall mean direction for the six flows groups at $D = 347.6°$, $I = 39.4°$, $\alpha_{95} = 27.5°$, $k = 7.1$ in geographic coordinates, and $D = 3.1°$, $I = 55.6°$, $\alpha_{95} = 9.8°$, $k = 47.9$ in stratigraphic coordinates. The

Table 4
F-test on Tertiary paleopoles

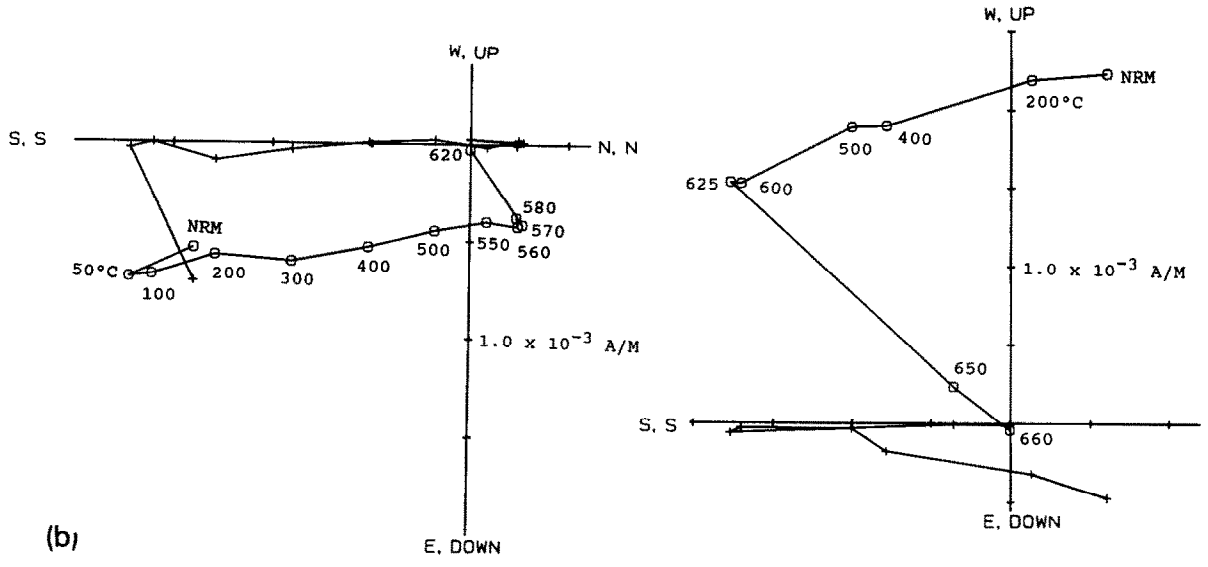
Studies	F-table (95%)	F-statistics	Interpretation	
Mongolian Block				
This study vs. Zheng et al. (1991)	F(2, 120)	> ~ 3.0	1.2	Indistinguishable
North China Block				
This study vs. Lin et al. (1985)	F(2, 28)	> ~ 3.3	4.5	Distinguishable
South China Block				
This study vs. Liang et al. (1986)	F(2, 114)	> ~ 3.0	2.0	Indistinguishable
Mongolian Block vs. NCB	F(2, 8)	> 4.5	1.1	Indistinguishable NCB vs. SCB
	F(2, 8)	> 4.5	0.3	Indistinguishable Mongolian Block vs. SCB
	F(2, 8)	> 4.5	1.3	Indistinguishable

N = samples or sites used in statistical analysis. See Table 1 for explanation of other abbreviations.
[a] Uncertainty estimated from a stereograph (with 28 samples from two localities) given in the original publication.
[b] Only one mean result was given in the original publication even though samples were from two localities.

tilt test is positive at the > 95% confidence level, as indicated by the factor of 6.2 times increase in Fisher's precision parameter, k, further suggesting that the characteristic magnetization is primary. The paleomagnetic results are presented in Tables 1 and 2.



Red sandstone samples



Yellow sandstone samples

Fig. 5. Representative vector plots of Tertiary sandstones from Guangxi. (a) Red sandstones from Wuxianling locality. (b) Yellow sandstones from Qingxiushan locality. Directions are plotted in geographic coordinates. Crosses and circles as Fig. 3.

3.2.2. Jining area

Like its counterpart in the Pingzhuang area, both AF and thermal demagnetization methods were equally successful for most of the Hannuoba basalts. In a small fraction of samples thermal demagnetization was necessary to remove a secondary overprint carried by hematite (Fig. 4b). The fact that the same characteristic direction was carried by both magnetite and hematite suggests that the hematite was probably formed by auto-oxidation processes during primary cooling of the basalts. Moreover, both normal and reversed polarities of the characteristic direction are again present in the stratigraphic sequence.

The regional generalized fold test for the two localities in Inner Mongolia is positive, demonstrating the prefolding age for the characteristic magnetization. As indicated in Table 2, the best estimate (K) of Fisher's precision parameter for the mean VGP position of the 13 basalt flows before and after tectonic correction is 9.9 and 26.3, respectively. Applying the classical fold test (McElhinny, 1964), the increase in K corresponds to a positive fold test at the $>99\%$ confidence level for the Tertiary basalt flow means. Due to these convincingly positive fold and reversal tests, the straightforward demagnetization behavior, and because the results are derived from widely separated localities, we believe that the characteristic component is an original thermal remanent magnetization (TRM), acquired when the basalts were cooling, and can be converted into a Miocene paleomagnetic pole.

3.3. Nanning (SCB)

The demagnetization behavior for the Fenghuangshan Formation is relatively straightforward. The magnetization of both yellowish and red sandstones is dominated by a very high-coercivity magnetic mineral, probably hematite. Consequently, progressive thermal demagnetization at a minimum of 7 levels was used to resolve characteristic components. A component with a direction aligned roughly along the present field direction at the sampling locality was typically removed up to demagnetization treatments of 500°C. Further thermal demagnetization revealed, in all samples, a high unblocking temperature component of magnetization with a demagnetization trajectory that converges toward the origin of vector plots (Fig. 5). The direction of this final high temperature component has dual polarity, and passes a tilt test (Tables 1 and 2). Thus, we interpret the characteristic component as representing a record of the paleomagnetic field close in age to the deposition of these Early Tertiary sandstones.

4. Comparison with other Tertiary paleomagnetic results from China

Tables 1–3 list the Tertiary paleopoles from this study for Mongolian and the North and South China Blocks and the associated statistical parameters, averaged from the virtual geomagnetic pole position for each locality. A few previously published Tertiary paleomagnetic results are available from Inner Mongolia, NCB and SCB, and have also been summarized in Table 3. In order to compare these paleopoles with those from this study, we present a brief evaluation.

Zheng et al. (1991) reported another set of Tertiary paleomagnetic data from Chifeng (42.0°N, 118.6°E), in Inner Mongolia, and Zhangjiakou (41.1°N, 114.7°E), in Hebei, which is derived from the same basalt formations reported in this study, but from different localities. Their data also pass reversal and regional fold tests and the corresponding pole (Long./Lat. = 192.8°E/83.4°N, $A_{95} = 6.2°$) is statistically indistinguishable from our Tertiary poles (see the F-test results in Table 4).

The only previously reported Tertiary pole for the NCB is derived from four basaltic lava flows from a single locality (36.6°N, 118.5°E) in Shandong Province (Lin et al., 1985). The pole (Long./ Lat. = 177.2°E/ 76.4°N, $A_{95} = 7.7°$) agrees fairly well with the coeval poles of this study from the same province (marginally different at the 95% confidence level according to the F-test in Table 4).

Two Tertiary poles have also been reported for South China (Lin et al., 1985; Liang et al., 1986). The pole reported by Lin et al. (1985) was

derived from three lava flows (~ 10 Ma) at a single locality from Zhejiang Province and has been discarded by the original authors (see Lin and Fuller, 1990) because one of three K/Ar ages (see Lin (1984), p. 165) is clearly anomalous (also see Lee et al. (1987), p. 3591 for a discussion).

The second reported Tertiary pole came from the Kunming area (25.0°N, 102.9°E) of Yunnan Province (Liang et al., 1986); it was derived from 28, early Tertiary red and yellowish sandstone samples and falls at 325.2°E, 84.4°N, with $A_{95} \approx 10°$ (estimated from a stereographic projection given in the original publication). Progressive thermal demagnetization was applied to resolve magnetic components. Both normal and reversed polarities of the characteristic magnetization were found in these redbeds. The primary nature of the characteristic magnetization is further supported by the apparently close correlation of polarity patterns with the geomagnetic polarity time scale for the late Eocene. Because the sampled localities are only about 150 km east of the Red River fault zone and the Tertiary strata in Yunnan can be stratigraphically correlated with the Tertiary Fenghuangshan Formation in Guangxi, we have compared the mean declinations of sampling sites in Yunnan and Guangxi to see if any paleomagnetically detectable rotations have occurred since the remanence was acquired. Although the Red River fault zone is believed to have experienced a great deal of left-lateral offset during the Tertiary (Tapponnier et al., 1990), we are surprised to see that the paleomagnetic data indicated only a small, statistically insignificant, counterclockwise rotation (9.7° ± 19.0°) between the two sampled areas.

The Tertiary poles for Inner Mongolia and North and South China are statistically indistinguishable (see Table 4 and Fig. 6). This is consistent with the hypothesis that Inner Mongolia and North and South China remained in the same relative position as today during the Tertiary, as discussed below.

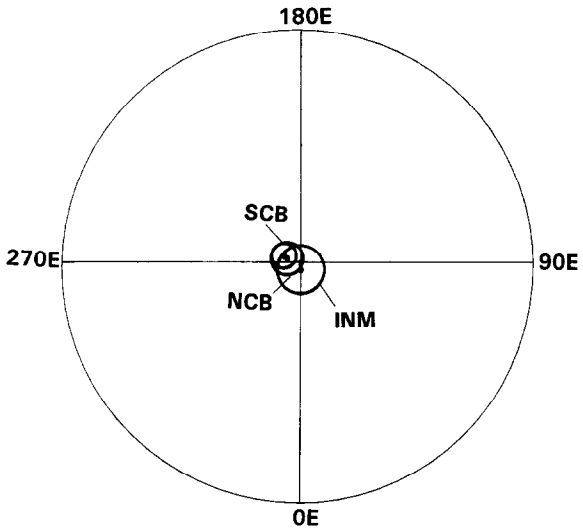


Fig. 6. Equal-area projection of Tertiary poles with 95% confidence limits for Inner Mongolia, North and South China Blocks (see the italicized data in Table 2). Abbreviations as Fig. 1.

5. Extrusion tectonics model and previous paleomagnetic interpretations

As stated previously, it has been suggested that many continental blocks in eastern Asia have achieved their present position through substantial tectonic displacement (rotation and/or translation) by the Cenozoic collision of India against Asia. Based on the resemblance between the geometry of great strike-slip faults now observed in Asia (e.g. Red River, Altyn Tagh, etc.) and that of faults in two-dimensional indentation experiments, many workers (Molnar and Tapponnier, 1975; Tapponnier et al., 1982; 1988; Peltzer and Tapponnier, 1988) have proposed a large-scale kinematic model of Cenozoic tectonics in Asia. According to this model, Siberia and other blocks in western Eurasia are considered as a buttress because the continental lithosphere in this region is assumed to offer more resistance to lateral motions than subduction zones along the Pacific and Indonesian margins. Thus, much of Indochina and China appears to have been pushed eastward to make way for the penetrating Indian continent. The model suggests that the Indochina block rotated about 25° clockwise and extruded about 800 km southeastward along the Red River fault, opening the South China Sea

Table 5
Selected paleomagnetic poles since the Late Permian for the major continental blocks of Asia

Age	Block	Long.	Lat.	A_{95}	N	Source and type of data
Upper Permian	EUR	155.0	51.6	7.6	4	Zhao et al. (1990a), EUR selection
		151.0	50.0	4.0	12	Frei and Cox (1987), 235–265 Ma
		153.0	50.8	8 [a]	2	Average of the above 2 studies
	NCB	358.3	47.8	7.7	9	Zhao et al. (1990a), average of 9 studies
	SCB	240.6	50.0	8.4	6	Zhao and Coe (1989), average of selected studies
		241.0	50.0	3.0	10	W. Fang et al. (1990)
		240.8	50	8 [a]	2	Average of the above 2 studies
Early Triassic	EUR	149.5	48.8	4.0	6	Zhao et al. (1990a)
		148.0	50.0	4.0	15	Frei and Cox (1987), 225–255 Ma
		148.8	49.4	5 [a]	2	Average of the above 2 studies
	NCB	358.1	57.3	8.6	5	Zhao et al. (1990a)
		352.8	57.7	6.3	3	Yang et al. (1991)
		355.5	57.5	5 [a]	2	Average of the above 2 studies
	SCB	217.7	45.7	4.5	10	Enkin et al. (1992) (not independent data)
		217.8	48.0	5.3	8	Zhao and Coe (1989)
Middle–Upper Triassic	EUR	137	59	8	?	Khramov et al. (1985), data from Siberia
		130	56	10	5	Frei and Cox (1987), 205–235 Ma
		140.8	53.5	3.4	19	Theveniaut et al. (1992), new data from France
		136.0	56.2	6.3	3	Average of the above 3 studies
	NCB	9.6	60.2	11.6	4	Enkin et al. (1992)
	SCB	194.4	45.1	16.8	5	Enkin et al. (1992)
Lower Jurassic	EUR	121.0	70.0	9	3	Westphal et al. (1986), transferred data from NA, AF and SA (not independent data, superseded by Besse and Courtillot, 1991)
		109.0	66.7	3.9	12	Besse and Courtillot (1991), transferred data from NA, AF and IND, and selection EUR, 180–200 Ma
	NCB	286.0	82.4	6.8	10	Yang et al. (1992)
	SCB	192.4	69.4	7.4 [b]	1	Enkin et al. (1992)
Middle Jurassic	EUR	150	71	8	4	Westphal et al. (1986), transferred data from 4 continents (superseded by Besse and Courtillot, 1991)
		120.7	63.3	3	16	Besse and Courtillot (1991), 160–180 Ma
	NCB	214	76.1	12.8	12	D.J. Fang et al. (1988)
		199.9	76.2	5.9	19	Zheng et al. (1991)

	SCB	186.2	65.0	15 [a]	2	Enkin et al. (1992)
Upper Jurassic	EUR	157.8	70.0	6.6	7	Besse and Courtillot (1991), 140–160 Ma (not indepedent data)
		180.0	71.0	16	3	Westphal et al. (1986), transferred data, 130 Ma
		206.5	69.9	7	4	Zhao et al. (1990a), average of transferred data from 4 studies
	NCB	231.6	68.5	9.5	4	Zhao et al. (1990a), data from Inner Mongolia
	SCB	235.3	74.6	13.2	5	Enkin et al. (1991a,b)
		213.7	73.0	12.6	6	Lin et al. (1985)
		224.0	74.1	14 [a]	2	Average of the above 2 studies
Lower Cretaceous	EUR	207.2	72.2	4.5	8	Zhao et al. (1990a) (not independent data, superseded by Besse and Courtillot, 1991)
		207.4	75.2	4	10	Besse and Courtillot (1991), transferred data, 95–130 Ma.
	NCB	210.8	75.9	7	10	Ma et al. (1991)
	SCB	204.6	69.0	4.3	7	Huang and Opdyke (1992)
		229.0	74.5	4.0	23	Enkin et al. (1991a,b)
		215.0	72.1	6 [a]	2	Average of the above 2 studies
Upper Cretaceous	EUR	188	78	12	3	Westphal et al. (1986), transferred data, 80 Ma (superseded by Besse and Courtillot, 1991)
		199.2	76.8	2.7	19	Besse and Courtillot (1991), 65–90 Ma.
	NCB	170.1	79.6	4.2	29	Zheng et al. (1991)
		249.5	82.9	5.7	6	Zhao et al. (1990a)
		201.0	83.2	15 [a]	2	Average of the above 2 studies
	SCB	220.9	81.5	7.1	18	Huang and Opdyke (1992)
		241.1	72.8	6.6	16	Enkin et al. (1991a,b)
		234.4	77.3	10 [a]	2	Average of the above 2 studies
Tertiary	EUR	144.7	81.2	2.4	5	Besse and Courtillot (1991), 10–50 Ma
		148	80	4	5	Achache et al. (1983), Miocene, west Europe
		167	83	5	4	Achache et al. (1983), Pliocene, west Europe
		174	79	6	14	Westphal and Durand (1989), British Tertiary province
		158.6	81.0	3.3	4	Average of the above 4 studies
	NCB	200.1	86.3	5.9	6	Average of this study, Zheng et al. (1991) and Lin et al. (1985)
	SCB	241.7	85.4	13.5	3	This study and Liang et al. (1986), see Tables 3 and 4

NA = North America, SA = South America, AFR = Africa, IND = India.
Age = approximate age of rocks studied, based mainly on palentological and geological evidence.
[a] Uncertainty for average of two poles estimated from studies' A_{95} and scatter.
[b] Uncertainty of a single pole.

(Peltzer and Tapponnier, 1988; Tapponnier et al., 1990). Next, the Altyn Tagh–Gansu–Weiho fault system, which extends from western Kunlun to eastern Qinling (total length in excess of 3500 km), was activated and became a major left-lateral strike-slip fault. The model predicts that, after the middle Eocene (about 15 Ma), South China and Tibet moved up to 1080 km eastward along this fault system (Peltzer and Tapponnier, 1988).

In contrast, some geological and geophysical studies argue against the extrusion/eastward displacement model (Dewey, 1988; England and Houseman, 1988; England and Molnar, 1990). For example, Dewey (1988) concludes that the Cenozoic tectonics of China can be separated geographically by the so-called 'South–North' seismic belt, which approximately follows the 100–105°E lines of longitude. West of this belt, the tectonic style is dominated by north–south shortening; east of this belt, the Cenozoic deformation of China is characterized by east–west extension, probably driven by western Pacific subduction.

Paleomagnetism can be used to test these different models and predictions. In recent years paleomagnetic data have been published on rocks from the Khorat plateau, Thailand (Barr et al., 1978; Bunopas, 1981; Maranate, 1982; Achache et al., 1983; Achache and Courtillot, 1985; Maranate and Vella, 1986; Chen and Courtillot, 1989; Yang and Besse, 1993). Although the age of magnetization is subject to different explanations, all paleomagnetic data are consistent in indicating about 25° of clockwise rotation of Thailand with respect to stable Eurasia since the Miocene. Hence, the paleomagnetic data from the Indochina Block appear to support large-scale clockwise rotation of this region during the first phase of collision, as predicted by the extrusion tectonics model.

Farther north, on the South China Block, Achache et al. (1983) concluded that the SCB (together with Indochina) has rotated about 10° clockwise since the late Tertiary. Interpretation of preliminary paleomagnetic APW paths by Lin et al. (1985) yielded an estimate of 4000 km relative motion between the South China Block and northern Eurasia, although no rotation was specified, mainly due to the Indian–Eurasian continental collision in the Cenozoic. If this latter conclusion were true, then the huge amount of displacement would require re-evaluation of any Mesozoic tectonic interpretations based on pre-Cenozoic paleomagnetic poles for the South China Block. We will see, however, that such displacements are not justified by the paleomagnetic data.

The question of to what extent the collision has controlled the Cenozoic tectonics of northern China and Siberia also remains a matter of debate. Achache et al. (1983) concluded that Eurasia underwent a continuous clockwise rotation of about 15° with respect to the spin axis from the Eocene to the present, although the authors did not mention whether it is related to the extrusion model. A recent study of Cretaceous redbeds from Tajikistan in the southern margin of Eurasia, by Pozzi and Feinberg (1991), however, revealed a total of $26° \pm 9°$ counterclockwise rotation of the sampled region subsequent to the India-Asia collision. Although the extrusion model does not predict large displacements and rotations for the North China Block and Tarim, differences in shape between the late Mesozoic paleomagnetic APW paths for Eurasia (Irving and Irving, 1982), North China (Lin et al., 1985), and Tarim (Li, 1990) have been interpreted as suggesting substantial relative motions between these blocks. Lin and Fuller (1986; 1990) concluded that the North China Block moved eastward relative to northern Eurasia at least 2500 km in the Cenozoic. Li (1990) suggested that Tarim and other amalgamated blocks have moved eastward about 2000 km relative to Siberia since the Cretaceous, and this motion is very likely due to the Cenozoic extrusion event. In contrast to Li's (1990) interpretation, and based on a comparison with the newly revised APW path for Eurasia by Besse and Courtillot (1991), Chen et al. (1991) concluded that both Tarim and Junggar were attached to Kazakhstan and Siberia prior to the Upper Jurassic. At a finer level of analysis (second-order variations), Chen et al. (1991) have also suggested that, in addition to about $5.9° \pm 3.7°$ continental shortening in the Altai Mountains, the Junggar basin has slightly rotated (8.6°

± 8.7°) counterclockwise with respect to the Tarim block, probably as a result of indentation by India.

Obviously, the relative displacements of Chinese blocks with respect to Eurasia due to the India–Asia collision need further testing. Because the number of Cenozoic paleomagnetic studies for these blocks is few and there is currently no Tertiary paleomagnetic data for Siberia, it is important to examine the late Mesozoic paleomagnetic data from Siberia to determine whether or not the large relative motions that have been inferred are a true tectonic feature in the Cenozoic tectonics of eastern Asia.

As will be discussed in the next section, a critical examination of late Mesozoic APW paths for Siberia and Eurasia does, indeed, show that many of the data are questionable and should be excluded for tectonic interpretation. The poles that we use for this purpose are given in Table 5.

6. Discussion of late Mesozoic paleomagnetic data

6.1. Siberia

Compared with Late Paleozoic data, the post-Early Triassic to Cenozoic paleomagnetic results from Siberia are scarce. A total of 19 Mesozoic poles were listed in a recent compilation (Piper, 1988), but many of these poles include data from orogenic areas or are based on data that have not been adequately demagnetized (Lee et al. (1987), p. 3592; see also Piper (1988), for those marked 'NT' = not treated). For these reasons, we do not use these poles for Siberia. Only two poles (one for the Late Jurassic and one for the Cretaceous) have been used in a paper on plate reconstructions by Khramov et al. (1985). Both poles are poorly defined and lack basic paleomagnetic information (e.g., number of samples, rock types, demagnetization work and stability tests). Therefore, we also omit these two poles from our tectonic interpretation.

As mentioned above, there is no reliable Tertiary pole for Siberia (Westphal et al., 1986, p. 42), which makes it difficult to uncover the magnetic history of older rocks and guard against the uncritical acceptance of possible Tertiary overprints for primary magnetizations.

6.2. Europe

The lack of reliable data of Jurassic and Cretaceous ages for Europe is a well-known problem (McElhinny, 1973, p. 244; Besse and Courtillot, 1991). Because of the lack of good outcrops with fold tests and the characteristic weakness of magnetization of sediments of these ages in tectonically stable areas of Europe (Westphal and Durand, 1989), many Jurassic–Cretaceous poles for Europe published earlier were derived from tectonically disturbed areas and may be significantly biased by incompletely removed overprints (Westphal et al., 1986; Lee et al., 1987; Westphal and Durand, 1989; Kim and Van der Voo, 1990). At present, only a few Cretaceous poles are available, derived mainly from Germany and France (Heller and Channell, 1979; Achache et al., 1983). Lower Tertiary poles are rather better determined because large amounts of paleomagnetic studies on the British Tertiary Igneous Province are available (Westphal and Durand, 1989).

Due to the fact that there are so few reliable European poles for the Mesozoic, an alternative way to estimate the Mesozoic pole positions for Europe is to transfer 'reliable' paleomagnetic data from other plates using reconstructions based on the identification of oceanic magnetic anomalies (Van der Voo, 1990). Using this approach, Westphal et al. (1986) and Besse and Courtillot (1991) transferred the North American APW path (Harrison and Lindh, 1982; May and Butler, 1986) and two selected African poles to Europe and Siberia, thereby obtaining 'transferred' Late Jurassic poles. Early Cretaceous transferred poles are more numerous than Late Jurassic poles and come from transferring paleomagnetic data from North and South America, India, and Africa (Van der Voo and French, 1974; Westphal et al., 1986; Besse and Courtillot, 1991).

We believe the transferred data are better for the Jurassic–Cretaceous for the following reasons. As mentioned above, several other workers in addition to ourselves have suggested that the Mesozoic Siberia and Europe data may be in

serious error (Lee et al., 1987, p. 3580; Frei and Cox, 1987; Kim and Van der Voo, 1990; Zhao et al., 1990a,b; Besse and Courtillot, 1991). On the other hand, the transferred poles for Europe agree well with the new Upper Cretaceous pole of Heller and Channell (1979) and, together, they give the best Jurassic–Cretaceous pole for Europe. We also note that the transferred Late Permian and Early Triassic poles (justified because the breakup of the North Atlantic did not start until the Early Cretaceous) agree well not only with each other (see Westphal et al., 1986; Van der Voo, 1990), but also with coeval poles from Europe and Siberia (Khramov et al., 1985; Theveniaut et al., 1992). For all these reasons, we will use the transferred data for the Late Jurassic and Cretaceous for Europe (Table 5).

6.3. Northern China (Inner Mongolia and North China Block)

We recently presented new paleomagnetic results on Mesozoic rocks from Inner Mongolia (Zhao et al., 1990a). The results of our study were derived from Late Jurassic and Cretaceous rock units in a wide area encompassing several geographic districts of Inner Mongolia. The data pass fold and reversal tests and are in good agreement with each other. Thus they are interpreted as representing a primary remanence.

Several Late Jurassic poles have been published (Lin et al., 1985; Yang, 1987; Cheng et al., 1988). In all studies, remanent directions are entirely of normal polarity and no documentation was offered to demonstrate that these rocks escaped the Cretaceous to Recent overprinting that is pervasive in eastern China (Zhao and Coe, 1984; Kent et al., 1987; Zhao, 1987). For this reason, we prefer not to consider these results in the final analysis until the ages of remanence acquisition are further constrained. D.J. Fang et al. (1988) recently reported a Middle–Late Jurassic pole for the NCB, which is derived from 12 samples (out of a total of 60 sampled) of the Middle–Late Jurassic Yungang Formation near Jingle (38.6°N, 112.1°E) in Shanxi Province. Two components of magnetization were evident in these sandstones. Thermal demagnetization revealed a characteristic component with both normal (75%) and reversed (25%) directions. The two polarities marginally pass the reversal test.

A large number of paleomagnetic results have been derived from Cretaceous rocks from the North China Block (Lin, 1984; Xiong et al., 1987; Cheng et al., 1988; X.D. Tan, 1988; Zheng et al., 1991; Ma et al., 1991) and are of mixed quality. Evidence for stability was presented in the studies of Ma et al. (1991) and Zheng et al. (1991). Data from these studies pass either the regional fold test and/or a conglomerate test, suggesting that the magnetization may be primary.

6.4. South China Block

Up to May 1990, when the first version of this paper was submitted, the Mesozoic–Cenozoic data base for the South China Block was still very sparse. However, 1991 saw a major increase in the number of Mesozoic paleomagnetic data for the South China Block, mainly through international cooperative efforts (Chan, 1991; Coe et al., 1991; Enkin et al., 1991a,b; Gilder et al., 1991; Wang and Van der Voo, 1991; Zhai et al., 1991; Dobson and Heller, 1992; Huang and Opdyke, 1992a). There is still a large variation in quality and reliability of the published paleomagnetic poles. In some cases, however, sampling sites have been re-studied, thus superseding former studies (e.g., Enkin et al., 1991a,b; Huang and Opdyke, 1992b).

Of the three published Late Jurassic poles for the South China Block (Lin et al., 1985; Zhu et al., 1988; Enkin et al., 1991b), the pole reported by Enkin et al. (1991b) gives the best determination. This pole came from the Feixianguan area (29.9°N, 102.9°E) of Sichuan Province, which was derived from 62 redbed samples. The data pass both reversal and fold tests. The remanent directions from which the other two Late Jurassic poles are derived (Zhu et al., 1988; Lin et al., 1985) are entirely of normal polarity and no fold or other field tests are available. The pole reported by Zhu et al. (1988) is also from Sichuan Province but is statistically different from the coeval pole by Enkin et al. (1991b) from the same province. Because no details concerning the age

of magnetization are given, we rejected the results by Zhu et al. (1988). However, a close examination of the data reported by Lin et al. (1985) revealed that the results were derived from three Late Jurassic formations from Zhejiang Province containing both igneous and sedimentary rocks. For this reason, and because the pole position falls very close to those of Enkin et al. (1991b), we include Lin et al.'s pole for the South China Block.

The sum total of Cretaceous paleomagnetic studies to date paints a reasonable picture of a high quality data set for the South China Block (see Enkin et al., 1992 for an updated view). Because many studies were performed on samples from a single locality or a small region, we generally take the best studies which demonstrate the age of magnetization with certainty. These studies include Huang and Opdyke (1992b) and Enkin et al. (1991a,b) (Table 5).

The selected late Mesozoic paleomagnetic poles for Europe/Siberia, northern China (NCB + INM) and the South China Block (Yangtze and Hunnan blocks), as well as the Tertiary poles, are summarized in Table 5 and plotted in Fig. 7. For the purpose of clarity we have also added the relatively well-defined Late Permian and Early Triassic paleomagnetic poles for these blocks (Frei and Cox, 1987; W. Fang and Van der Voo, 1990; Zhao et al., 1990a; Enkin et al., 1992; Theveniaut et al., 1992).

The resulting APW paths for Europe/Siberia and North and South China form a striking 'three-legged' starfish, which summarizes much of the first-order relative motions of these three blocks since the Mesozoic. The tectonic implications for the pre-Late Jurassic period of this figure are discussed in detail by Opdyke et al. (1986), Zhao and Coe (1987; 1989), Zhao et al. (1990a), and Enkin et al. (1992). These discussions will not be repeated here since we are concerned with the post-Cretaceous relative motions of Chinese blocks with respect to Europe/Siberia in this study. As shown in Fig. 7, the three paths converge during the Late Jurassic and are tightly grouped throughout the Cretaceous and Tertiary. Therefore, it is clear from the APW paths that there has been no paleomagnetically detectable relative movement between these continental blocks of Asia since the Tertiary. We suggest that the inference of large-scale eastward displacement is erroneous, because it is based on comparisons with unreliable Jurassic and Cretaceous paleomagnetic data from Europe, rather

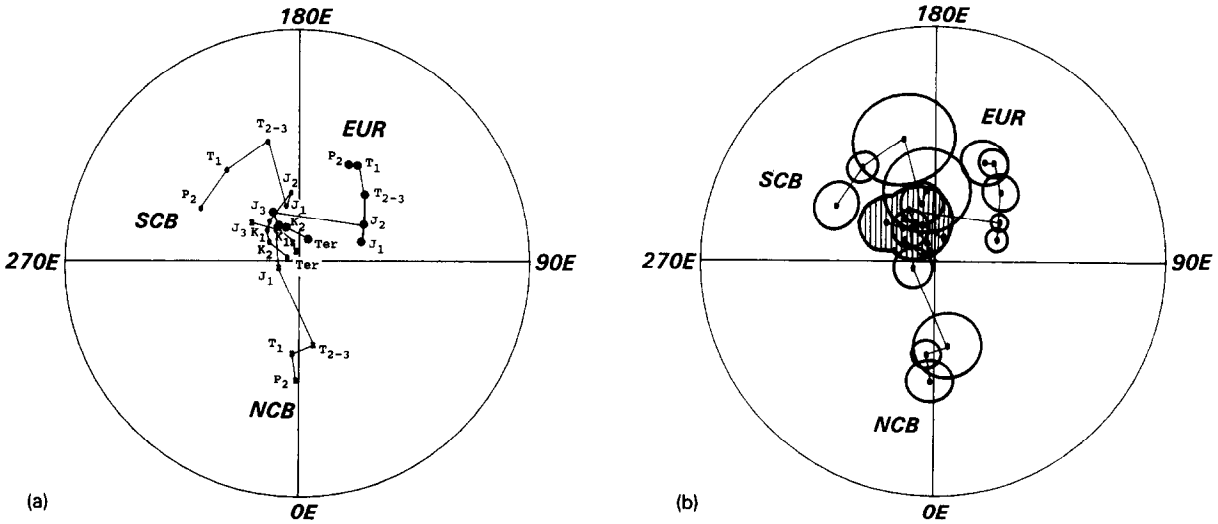


Fig. 7. Apparent polar wander paths since the Late Permian for Europe/Siberia (*EUR*), North and South China Blocks (*NCB* and *SCB*, respectively) (a) without and (b) with circles of 95% confidence (see Table 5 for underlined data). Shading = the 95% confidence limits for Late Jurassic poles.

than on the more reliable data transferred from other continents. For example, the Late Cretaceous pole (80 Ma) in the classic Eurasian APW path of Irving and Irving (1982) falls at 74°N, 145°E, while the newly established coeval poles cluster at 77°N, 189°E (e.g., 83 Ma pole of Heller and Channell (1979), at 76°N/181°E; 80 Ma pole of Westphal et al. (1986), at 78°N/188°E; and 80 Ma pole of Besse and Courtillot (1991), at 76°N/199°E).

In summary, we find concordant paleomagnetic poles for the Tertiary from Inner Mongolia and North and South China. We postulate that small, second-order Cenozoic movements (less than 500 km, which is below the limit of accuracy of most paleomagnetic studies) between these Chinese blocks and Siberia may have occurred. These subtle second-order movements are much more difficult to resolve and certainly cannot be demonstrated by the paleomagnetic APW path analysis. The Cenozoic extrusion model (Tapponnier et al., 1982) elegantly accounts for the shortening and rotation in Tibet, Tarim and Junggar of western China, resulting from the India–Asia collision, and probably also for the current position of Indochina by southeastward displacement. The North and South China blocks, however, appear to have remained relatively unharmed by the India–Asia collision. Our paleomagnetic data imply that the free boundary condition in eastern China assumed by this model may need to be reconsidered; that is, the westward subduction of the Philippine Sea and Pacific plates to the east may provide substantial resistance to eastward extrusion of continental crust. The fact that no large eastward displacement seems to have occurred between North and South China and Siberia since the India–Asia collision may serve as one such line of evidence.

7. Conclusions

The new Tertiary paleomagnetic data from this study are consistent with the hypothesis that the North China Block, South China Block and Inner Mongolia were in the same position as today during the Tertiary. An analysis of Mesozoic paleomagnetic data corroborates this hypothesis. Most previously existing Jurassic and Cretaceous paleomagnetic poles from Europe and Siberia are of undemonstrated reliability and should not be used for tectonic reconstruction. By using poles derived mainly by transferring reliable data from other continents for this period, we show that the previous paleomagnetic inferences of large eastward displacements between these Chinese blocks and Siberia during the Cenozoic are no longer warranted. Nevertheless, the body of reliable paleomagnetic data is still small. Additional Mesozoic and Cenozoic paleomagnetic data from Siberia and North and South China are needed to test these conclusions further and to resolve the smaller, second-order displacements that appear to characterize the Cenozoic tectonic history of this part of Asia.

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